FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

2

RULE – BASED AGENT

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Date:	10/12/26

I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

II. TECHNICAL ASSESSMENT

RULE – BASED AGENT

Objective

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

Materials Needed

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
   (Example: TA2_Patilano.pdf)
8. Submit the **python file** and **PDF** file.

Laboratory Task

Pre-Lab Preparation

1. Review the Python concepts:
   - Classes and objects

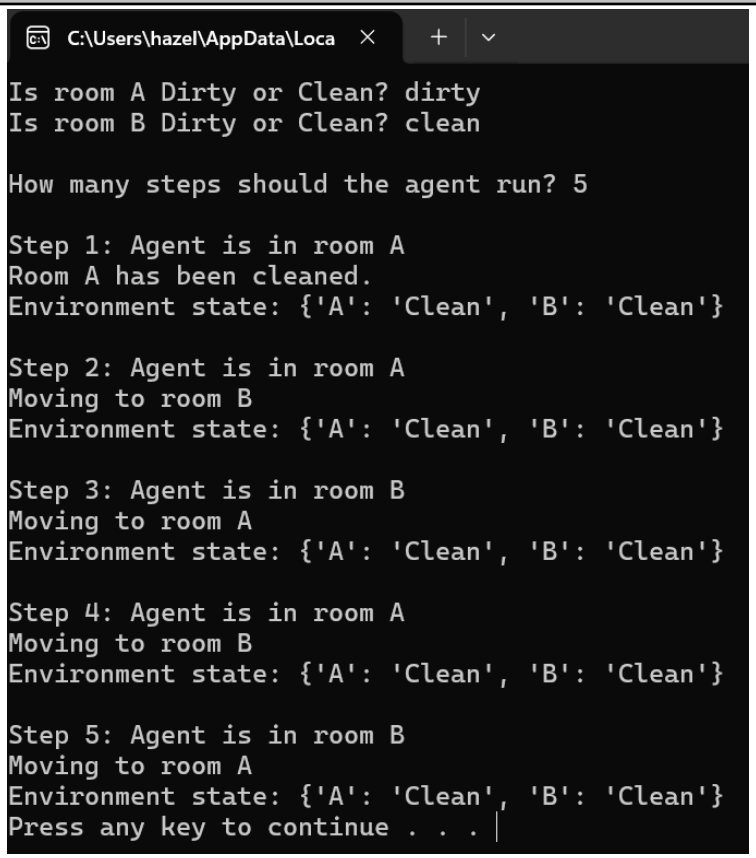
- If-Else statements
- Loops (for loop)
- Functions / methods

2. Understand the environment:
   - Two rooms: A and B
   - Each room can be Dirty or Clean

Task 1: Set Up the Environment

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output



Task 2: Implement the Rule-Based Agent

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:
   - Rule 1: If the current room is dirty, clean it.
   - Rule 2: If the current room is clean, move to the other room.
3. Implement a **move method** to switch rooms.

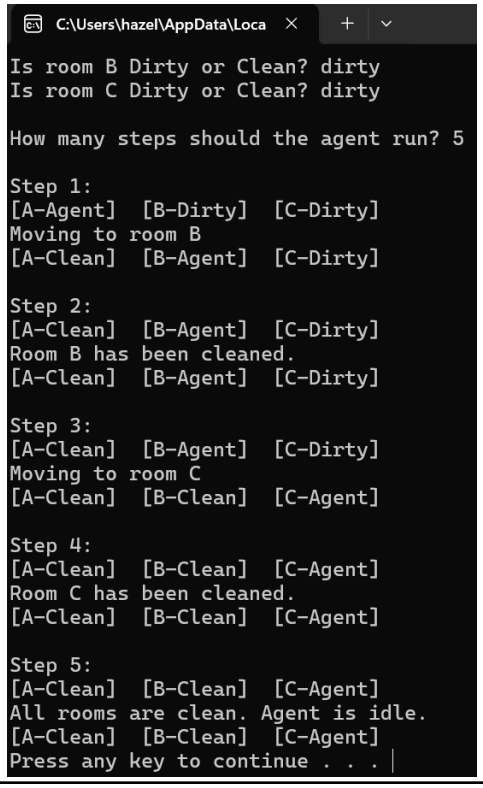
Task 3: Run the Simulation

1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.
3. After each step, print:
    - The agent's current location
    - The action taken (cleaned/moved)
    - Current state of the environment

Optional (Bonus Task)

- Add a **third room (C)** and update the agent logic.
- Make the agent **randomly choose which dirty room to clean next**.
- Visualize the agent moving between rooms using a **text-based grid**.

Sample Output (Bonus Task)



Your Code - Laboratory Tasks

```
def get_room_state(room_name):
    """Ask for and validate the initial state of one room."""
    while True:
        state = input(f"Enter the initial state of Room {room_name} (dirty/clean):
").strip().lower()
        if state in {"dirty", "clean"}:
            return state
        print("Invalid input. Please enter only 'dirty' or 'clean'.")


def is_dirty(environment, room):
    """Return True when the selected room is dirty."""
    return environment[room] == "dirty"


def clean_room(environment, room):
    """Change the selected room's state to clean."""
    environment[room] = "clean"


class VacuumAgent:
    """A rule-based agent that cleans or moves according to room state."""

    def __init__(self, environment, start_room="A"):
        self.environment = environment
        self.current_room = start_room

    def move(self):
        """Move between the two rooms."""
        self.current_room = "B" if self.current_room == "A" else "A"

    def perceive_and_act(self):
        """Perceive the current room and apply the first matching rule."""
        if is_dirty(self.environment, self.current_room):
            clean_room(self.environment, self.current_room)
            return f"cleaned Room {self.current_room}"

        previous_room = self.current_room
        self.move()
        return f"moved from Room {previous_room} to Room {self.current_room}"


def display_environment(environment):
    """Return a readable representation of the environment."""
    return f"A = {environment['A']}, B = {environment['B']}"


def get_step_count():
    """Ask for a non-negative integer number of simulation steps."""
    while True:
        try:
            steps = int(input("How many simulation steps should run? "))
            if steps >= 0:
                return steps
```

```
        except ValueError:
            pass
        print("Invalid input. Please enter a non-negative whole number.")


def main():
    print("RULE-BASED VACUUM AGENT")
    print("The agent starts in Room A.\n")

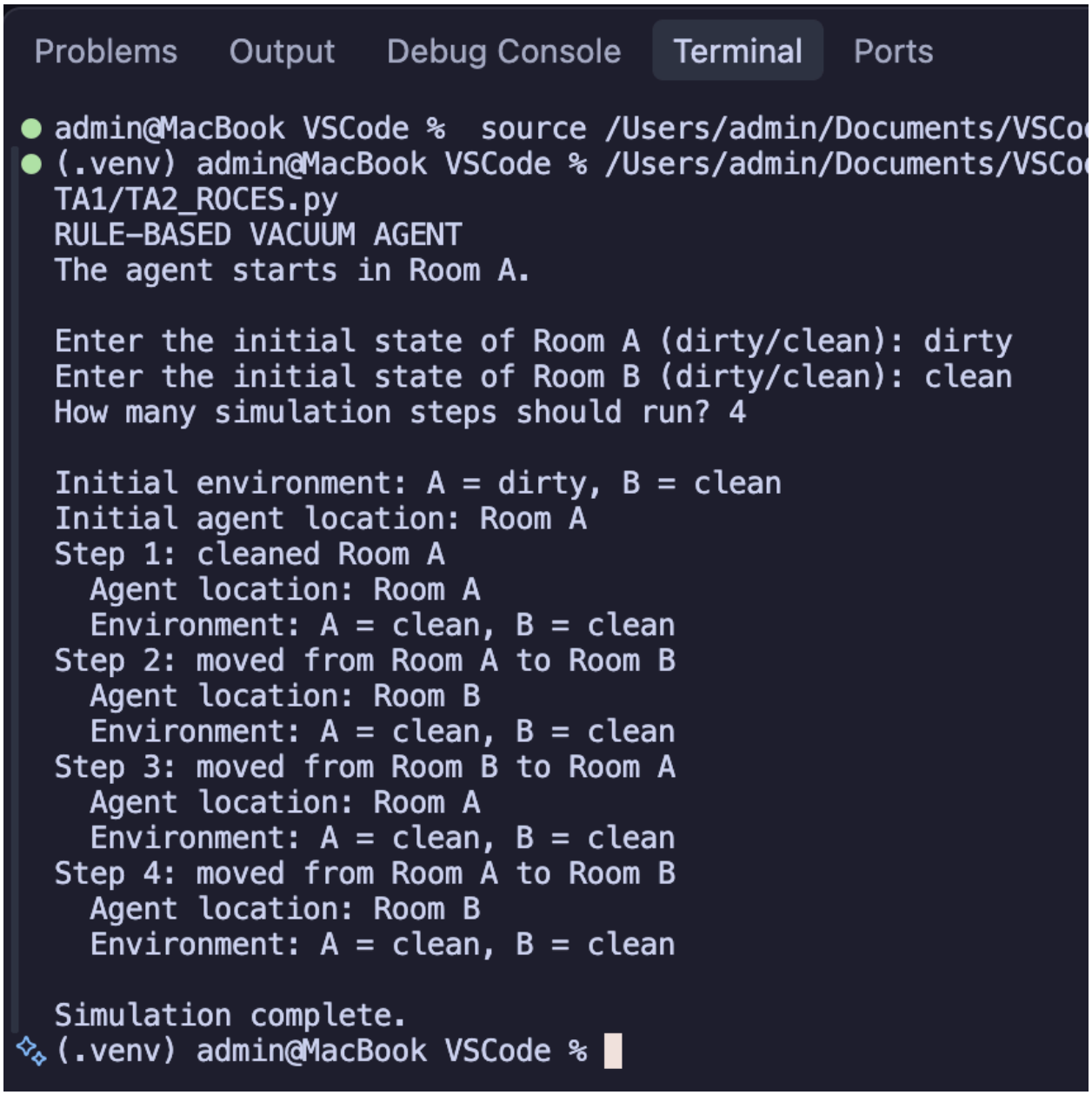
    environment = {
        "A": get_room_state("A"),
        "B": get_room_state("B"),
    }
    steps = get_step_count()
    agent = VacuumAgent(environment)

    print(f"\nInitial environment: {display_environment(environment)}")
    print(f"Initial agent location: Room {agent.current_room}")

    for step in range(1, steps + 1):
        action = agent.perceive_and_act()
        print(f"Step {step}: {action}")
        print(f"  Agent location: Room {agent.current_room}")
        print(f"  Environment: {display_environment(environment)}")

    print("\nSimulation complete.")


if __name__ == "__main__":
    main()
```


Problems
Output
Debug Console
Terminal
Ports
admin@MacBook VSCode % source /Users/admin/Documents/VSCo
(.venv) admin@MacBook VSCode % /Users/admin/Documents/VSCo
TA1/TA2_ROCES.py
RULE-BASED VACUUM AGENT
The agent starts in Room A.
Enter the initial state of Room A (dirty/clean): dirty
Enter the initial state of Room B (dirty/clean): clean
How many simulation steps should run? 4
Initial environment: A = dirty, B = clean
Initial agent location: Room A
Step 1: cleaned Room A
Agent location: Room A
Environment: A = clean, B = clean
Step 2: moved from Room A to Room B
Agent location: Room B
Environment: A = clean, B = clean
Step 3: moved from Room B to Room A
Agent location: Room A
Environment: A = clean, B = clean
Step 4: moved from Room A to Room B
Agent location: Room B
Environment: A = clean, B = clean
Simulation complete.
(.venv) admin@MacBook VSCode %

Task 4: Analysis Questions

After running the simulation, answer the following:

1. What rules does the agent follow?
2. How does the agent decide when to clean and when to move?
3. What happens if both rooms are already clean at the start?
4. How would you modify the agent to handle **more than two rooms**?

Answer the Questions ⇩

1. The agent follows two rules: first, if its current room is dirty, it cleans that room. Second, if its current room is clean, it moves to the other room.
2. The agent perceives the state of its current room. A dirty state triggers the clean action. A clean state triggers the move action.
3. The agent never needs to clean. On every step, it moves to the other room, alternating between Room A and Room B.
4. I would store all rooms in the environment dictionary and replace the two-room move logic with a list of room names. The agent could move to the next room in the list, or select a dirty room using a search or random-choice strategy. The perception and cleaning rules could remain the same.